

ANSWERS

1.

- (a) (i) oscillations are damped/amplitude decreases B1
as magnet moves, flux is cut by coil B1
e.m.f./current is induced in the coil B1
causing energy loss in load OR force on magnet B1
energy is derived from oscillations of magnet
OR force opposes motion of magnet B1
- (ii) $T = 0.60 \text{ s}$ C1
 $\omega_0 (= 2\pi/T) = 10.5 \text{ rad s}^{-1}$ A1
- (b) sketch: sinusoidal wave with period unchanged or slightly smaller M1
same initial displacement, less damping A1
- (c) (i) sketch: general shape – peaked curve M1
peak at ω_0 and amplitude never zero A1
- (ii) resonance B1
- (iii) useful: e.g. child on swing, microwave oven heating B1
avoid: e.g. vibrating panels, vibrating bridges B1
(for credit, stated example must be put in context)

2.

- (a) (i) $BI \sin \theta$ B1
- (ii) (downwards) into (the plane of) the paper B1
- (b) (i) magnetic field (due to current) in one loop OR each loop acts as a coil B1
cuts/is normal to current in second loop OR produces magnetic field B1
causing force on second loop OR fields in same direction M1
either Newton's 3rd discussed
or vice versa clear gives rise to attraction OR so attracts A1
- (ii) $B = 2 \times 10^{-7} I / 0.75 \times 10^{-2} (= 2.67 \times 10^{-5} I)$ C1
force = $0.26 \times 10^{-3} \times 9.81 (= 2.55 \times 10^{-3} \text{ N})$ C1
 $F = BIL$
 $2.55 \times 10^{-3} = 2.67 \times 10^{-5} \times I^2 \times 2\pi \times 4.7 \times 10^{-2}$ C1
 $I = 18 \text{ A}$ A1

3.

(a) parallel (to the field) B1

(b) (i) torque = $F \times d$
 $2.1 \times 10^{-3} = F \times 2.8 \times 10^{-2}$ C1
 $F = 0.075 \text{ N}$ A1
(use of 4.5 cm scores no marks)

(ii) zero A1

(c) $F = BILN(\sin\theta)$ C1
 $0.075 = B \times 0.170 \times 4.5 \times 10^{-2} \times 140$ M1
 $B = 7.0 \times 10^{-2} \text{ T} = 70 \text{ mT}$ A0

(d) (i) (induced) e.m.f. is proportional to / equal to rate of change of M1
(magnetic) flux (linkage) A1

(ii) change in flux linkage = BAN
 $= 0.070 \times 4.5 \times 10^{-2} \times 2.8 \times 10^{-2} \times 140$ C1
 $= 0.0123 \text{ Wb turns}$
induced e.m.f = $0.0123 / 0.14$ C1
 $= 88 \text{ mV}$ A1

(Note: This is a simplified treatment. A full treatment would involve the averaging of $B \cos\theta$ leading to a $\sqrt{2}$ factor)

4.

(a) region (of space) / area where B1
a force is experienced by M1
current-carrying conductor / moving charge / permanent magnet A1

(b) (i) electric B1

(ii) gravitational B1

(iii) magnetic B1

(iv) magnetic B1

5.

- (a) unit of magnetic flux density / magnetic field strength B1
 (uniform) field normal to wire carrying current of 1 A M1
 giving force (per unit length) of 1 N m^{-1} A1
- (b) (i) force on magnet / balance is downwards (so by Newton's third law) B1
 force on wire is upwards M1
 pole P is a north pole A1
- (ii) $F = BIL$ and $F = mg$ (g missing, then 0/3 in (ii)) C1
 $2.3 \times 10^{-3} \times 9.8 = B \times 2.6 \times 4.4 \times 10^{-2}$ ($g = 10$, loses this mark) C1
 $B = 0.20 \text{ T}$ A1
- (c) reading for maximum current = $2.3 \times \sqrt{2}$ C1
 total variation = $2 \times 2.3 \times \sqrt{2}$
 = 6.5 g A1

6.

- coil in series with meter (*do not allow inclusion of a cell*) B1
 push known pole into coil B1
 observe current direction (*not reading*) B1
 (induced) field / field from coil repels magnet B1
either states rule to determine direction of magnetic field in coil
or reversing magnet direction gives opposite deflection on meter B1
 direction of induced current such as to oppose the change producing it B1

7.

- (a) concentric circles ... (*at least three lines*) M1
 with increasing separation A1
 correct direction clear B1
- (b) (i) correct position to left of wire B1
- (ii) $B = (4\pi \times 10^{-7} \times 1.7) / (2\pi \times 1.9 \times 10^{-2})$ C1
 = $1.8 \times 10^{-5} \text{ T}$ A1
- (c) distance $\propto$ current C1
 current = $(2.8 / 1.9) \times 1.7$
 = 2.5 A A1

8.

- (a) (i) concentric circles, anticlockwise(minimum 3 circles)M1
 separation of lines increases with distance from wire A1
- (ii) direction from Y towards X A1
- (b) (i) flux density at wire Y = $(4\pi \times 10^{-7} \times 5.0) / (2\pi \times 2.5 \times 10^{-2})$ C1
 = $4.0 \times 10^{-5} \text{ T}$ C1
 force per unit length = BI
 = $4.0 \times 10^{-5} \times 7.0$ C1
 = $2.8 \times 10^{-4} \text{ N}$ A1
- (ii) either force depends on product of the currents in the two wiresM1
 so equal A1
 or (isolated system so) Newton's 3rd law applies (M1)
 so equal (A1)

9.

- (a) (i) V_H depends on angle between (plane of) probe and B -field B1
 either V_H max when plane and B -field are normal to each other
 or V_H zero when plane and B -field are parallel
 or V_H depends on sine of angle between plane and B -field B1
- (ii) 1 calculates $V_H r$ at least three times M1
 to 1 s.f. constant so valid or approx constant so valid
 or to 2 s.f., not constant so invalid A1
- 2 straight line passes through origin B1
- (b) (i) e.m.f. induced is proportional / equal to M1
 rate of change of (magnetic) flux (linkage) A1
 constant field in coil / flux (linkage) of coil does not change B1
- (ii) e.g. vary current (in wire) / switch current on or off / use a.c. current
 rotate coil
 move coil towards / away from wire (1 mark each, max 3) B3

10.

- (a) arrow pointing up the page B1
- (b) (i) $Eq = Bqv$ C1
 $v = (12 \times 10^3) / (930 \times 10^{-6})$ C1
 $= 1.3 \times 10^7 \text{ m s}^{-1}$ A1
- (ii) $Bqv = mv^2 / r$ C1
 $q/m = (1.3 \times 10^7) / (7.9 \times 10^{-2} \times 930 \times 10^{-6})$ C1
 $= 1.8 \times 10^{11} \text{ C kg}^{-1}$ A1

11.

- (a) (i) straight line with positive gradient through origin M1
A1
- (ii) maximum force shown at $\theta = 90^\circ$ M1
zero force shown at $\theta = 0^\circ$ M1
reasonable curve with F about $\frac{1}{2}$ max at 30° A1
- (b) (i) force on electron due to magnetic field B1
force on electron normal to magnetic field and direction of electron B1
- (ii) quote / mention of (Fleming's) left hand rule M1
electron moves towards QR A1

12.

- (a) field into (the plane of) the paper B1
- (b) force due to magnetic field provides the centripetal force B1
 $mv^2 / r = Bqv$ C1
 $B = (20 \times 1.66 \times 10^{-27} \times 1.40 \times 10^5) / (1.6 \times 10^{-19} \times 6.4 \times 10^{-2})$ B1
 $= 0.454 \text{ T}$ A0
- (c) (i) semicircle with diameter greater than 12.8 cm B1
- (ii) new flux density $= \frac{22}{20} \times 0.454$ C1
 $B = 0.499 \text{ T}$ A1

13.

- (a) magnetic flux $= BA$
 $= 89 \times 10^{-3} \times 5.0 \times 10^{-2} \times 2.4 \times 10^{-2}$ C1
 $= 1.07 \times 10^{-4} \text{ Wb}$ A1
- (b) (i) e.m.f. $= \Delta\phi / \Delta t$ C1
(for $\Delta\phi = 1.07 \times 10^{-4} \text{ Wb}$), $\Delta t = 2.4 \times 10^{-2} / 1.8 = 1.33 \times 10^{-2} \text{ s}$ C1
e.m.f. $= (1.07 \times 10^{-4}) / (1.33 \times 10^{-2})$
 $= 8.0 \times 10^{-3} \text{ V}$ A1
- (ii) current $= 8.0 \times 10^{-3} / 0.12$ M1
 $\approx 70 \text{ mA}$ A0
- (c) force on wire $= BIL$
 $= 89 \times 10^{-3} \times 70 \times 10^{-3} \times 5.0 \times 10^{-2}$ C1
 $\approx 3 \times 10^{-4} \text{ (N)}$ M1
suitable comment e.g. this force is too / very small (to be felt) A1

14.

- (a) force due to E -field is equal and opposite to force due to B -field B1
 $Eq = Bqv$ B1
 $v = E/B$ B1
- (b) *either* charge and mass are not involved in the equation in (a)
or F_E and F_B are both doubled
or E , B and v do not change M1
so no deviation A1

15.

- (a) region (of space) where there is a force M1
either on / produced by magnetic pole
or on / produced by current carrying conductor / moving charge A1
- (b) (i) force on particle is (always) normal to velocity / direction of travel B1
speed of particle is constant B1
- (ii) magnetic force provides the centripetal force B1
 $mv^2 / r = Bqv$ M1
 $r = mv / Bq$ A0

- (c) (i) direction from 'bottom to top' of diagram B1
- (ii) radius proportional to momentum C1
 ratio = $5.7 / 7.4$
 = 0.77 A1
(answer must be consistent with direction given in (c)(i))

16.

- (a) (i) (induced) e.m.f. proportional to M1
 rate of change of (magnetic) flux (linkage) / rate of flux cutting A1
- (ii) 1. moving magnet causes change of flux linkage B1
 2. speed of magnet varies so varying rate of change of flux B1
 3. magnet changes direction of motion (so current changes direction) B1
- (b) period = 0.75 s C1
 frequency = 1.33 Hz A1
- (c) graph: smooth correctly shaped curve with peak at f_0 M1
 A never zero A1
- (d) (i) resonance B1
- (ii) e.g. quartz crystal for timing / production of ultrasound A1

17.

- (a) (i) $Bqv(\sin\theta)$ or $Bqv(\cos\theta)$ B1
- (ii) qE B1
- (b) F_B must be opposite in direction to F_E B1
 so magnetic field into plane of paper B1

18.

(a) unit of magnetic flux density B1
 field normal to (straight) conductor carrying current of 1 A M1
 force per unit length is 1 N m^{-1} A1

(b) (i) force on particle always normal to direction of motion M1
 (and speed of particle is constant)
 magnetic force provides the centripetal force A1

(ii) $mv^2/r = Bqv$ M1
 $r = mv/Bq$ A0

(c) (i) the momentum/speed is becoming less M1
 so the radius is becoming smaller A1

(ii) 1. spirals are in opposite directions M1
 so oppositely charged A1

2. equal initial radii M1
 so equal (initial) speeds A1

19.

(a) sketch: concentric circles (*minimum of 3 circles*) M1
 separation increasing with distance from wire A1
 correct direction B1

(b) (i) arrow direction from wire B towards wire A B1

(ii) *either* reference to Newton's third law
or force on each wire proportional to product of the two currents M1
 so forces are equal A1

(c) force always towards wire A/always in same direction B1
 varies from zero (to a maximum value) (1)
 variation is sinusoidal / $\sin^2$ (1)
 (at) twice frequency of current (1)
 (*any two, one each*) B2

20.

- (a) (long) straight conductor carrying current of 1 A
current/wire normal to magnetic field
(for flux density 1 T,) force per unit length is 1 N m^{-1} M'
M'
A1
- (b) (i) (originally) downward force on magnet (due to current) B1
by Newton's third law (allow "N3") M'
upward force on wire A1
- (ii) $F = BIL$
 $2.4 \times 10^{-3} \times 9.8 = B \times 5.6 \times 6.4 \times 10^{-2}$ C1
 $B = 0.066 \text{ T}$ (need 2 SF) A1
(g missing scores 0/2, but g = 10 leading to 0.067T scores 1/2)
- (c) new reading is $2.4\sqrt{2} \text{ g}$ C1
either changes between +3.4 g and -3.4 g
or total change is 6.8 g A1

21.

- (a) (i) particle must be moving M1
with component of velocity normal to magnetic field A1
- (ii) $F = Bqv \sin \theta$ M1
 q , v and θ explained A1
- (b) (i) face BCGF shaded A1
- (ii) between face BCGF and face ADHE A1
- (c) potential difference gives rise to an electric field M1
either $F_E = qE$ (no need to explain symbols)
or electric field gives rise to force (on an electron) A1

22.

- (a) force on proton is normal to velocity and field M1
provides centripetal force (for circular motion) A1
- (b) magnetic force = Bqv B1
centripetal force = $m r \omega^2$ or $m v^2 / r$ B1
 $v = r \omega$ B1
 $Bqv = Bqr \omega = m r \omega^2$
 $\omega = Bq / m$ A1

23.

- (a) either $\phi = BA \sin \theta$ M1
 where A is the area (through which flux passes)
 θ is the angle between B and (plane of) A A1
 or
 $\phi = BA$ (M1)
 where A is area normal to B (A1)
- (b) graph: V_H constant and non zero between the poles and zero outside M1
 sharp increase/decrease at ends of magnet A1
- (c) (i) (induced) e.m.f. proportional to M1
 rate of change of (magnetic) flux (linkage) A1
- (ii) short pulse on entering and on leaving region between poles M1
 pulses approximately the same shape but opposite polarities A1
 e.m.f. zero between poles and outside A1

24.

- (a) (uniform magnetic) flux normal to long (straight) wire carrying a current of 1 A M1
 (creates) force per unit length of 1 N m^{-1} A1
- (b) (i) flux density $= 4\pi \times 10^{-7} \times 1.5 \times 10^3 \times 3.5$ C1
 $= 6.6 \times 10^{-3} \text{ T}$ A1
- (ii) flux linkage $= 6.6 \times 10^{-3} \times 28 \times 10^{-4} \times 160$ C1
 $= 3.0 \times 10^{-3} \text{ Wb}$ A1
- (c) (i) (induced) e.m.f. proportional to rate of M1
 change of (magnetic) flux (linkage) A1
- (ii) e.m.f. $= (2 \times 3.0 \times 10^{-3}) / 0.80$ C1
 $= 7.4 \times 10^{-3} \text{ V}$ A1

25.

- (a) (uniform magnetic) flux normal to long (straight) wire carrying a current of 1 A M1
 (creates) force per unit length of 1 N m^{-1} A1
- (b) (i) sketch: concentric circles M1
 increasing separation (*must show more than 3 circles*) A1
 correct direction (anticlockwise, looking down) B1

(ii)	$B = (4\pi \times 10^{-7} \times 6.3) / (2\pi \times 4.5 \times 10^{-2})$ $= 2.8 \times 10^{-5} \text{ T}$	C1 A1
(iii)	$F = BIL (\sin \theta)$ $= 2.8 \times 10^{-5} \times 9.3 \times 1$ $F/L = 2.6 \times 10^{-4} \text{ N m}^{-1}$	C1 A1
(c)	force per unit length depends on product $I_X I_Y$ / by Newton's third law / action and reaction are equal and opposite so same for both	M1 A1
26.		
(a) (i)	field shown as right to left	B1
(ii)	lines are more spaced out at ends	B1
(b)	Hall voltage depends on angle <i>either</i> between field and plane of probe <i>or</i> maximum when field normal to plane of probe <i>or</i> zero when field parallel to plane of probe	M1 A1
(c) (i)	(induced) e.m.f. proportional to rate of change of (magnetic) flux (linkage) (<i>allow rate of cutting of flux</i>)	M1 A1
(ii)	e.g. move coil towards/away from solenoid rotate coil vary current in solenoid insert iron core into solenoid (<i>any three sensible suggestions, 1 each</i>)	B3
27.		
(a)	force due to magnetic field is constant force is (always) normal to direction of motion this force provides the centripetal force	B1 A1
(b)	$mv^2 / r = Bqv$ hence $q / m = v / Br$	M1 A0
(c) (i)	$q / m = (2.0 \times 10^7) / (2.5 \times 10^{-3} \times 4.5 \times 10^{-2})$ $= 1.8 \times 10^{11} \text{ C kg}^{-1}$	C1 A1
(ii)	sketch: curved path, constant radius, in direction towards bottom of page tangent to curved path on entering and on leaving the field	M1 A1

28.

- (a) (i) region (of space)
either where a moving charge (may) experience a force
or around a magnet where another magnet experiences a force B1
- (ii) $(\Phi =) BA \sin \theta$ A1
- (b) (i) plane of frame is always parallel to B_v /flux linkage always zero B1
- (ii) $\Delta \Phi = 1.8 \times 10^{-5} \times 52 \times 10^{-2} \times 95 \times 10^{-2}$ C1
 $= 8.9 \times 10^{-6} \text{ Wb}$ A1
- (c) (i) (induced) e.m.f. proportional to rate of change of (magnetic) flux (linkage) M1
(allow rate of cutting of flux) A1
- (ii) e.m.f. $= (8.9 \times 10^{-6}) / 0.30$
 $= 3.0 \times 10^{-5} \text{ V}$ A1

29.

- (a) *either* constant speed parallel to plate
or accelerated motion/force normal to plate/in direction field B1
 so not circular A0
- (b) (i) direction of force due to magnetic field opposite to that due to electric field B1
 magnetic field into plane of page B1
- (ii) force due to magnetic field = force due to electric field B1
 $Bqv = qE$
 $B = E / v$ C1
 $= (2.8 \times 10^4) / (4.7 \times 10^5)$
 $= 6.0 \times 10^{-2} \text{ T}$ A1
- (c) (i) no change/not deviated B1
- (ii) deviated upwards B1
- (iii) no change/not deviated B1

30.

- (a) graph: V_H increases from zero when current switched on B1
 V_H then non-zero constant B1
 V_H returns to zero when current switched off B1
- (b) (i) (induced) e.m.f. proportional to rate M1
of change of (magnetic) flux (linkage) A1
- (ii) pulse as current is being switched on B1
zero e.m.f. when current in coil B1
pulse in opposite direction when switching off B1

31.

- (a) only curve with decreasing gradient M1
acceptable value near $x = 0$ and does not reach zero A1

(if graph line less than 4.0 cm do not allow A1 mark)
(no credit if graph line has positive and negative values of V_H)

- (b) graph: from 0 to 2T, two cycles of a sinusoidal wave M1
all peaks above 3.5 mV C1
peaks at 4.95/5.0 mV (*allow 4.8 mV to 5.2 mV*) A1

- (c) e.m.f. induced in coil when magnetic field / flux is changing / cutting B1
- either* at each position, magnetic field does not vary
so no e.m.f. is induced in the coil / no reading on the millivoltmeter
or at each position, switch off current and take millivoltmeter reading
or at each position, rapidly remove coil from field and take meter reading B1

32.

- (a) electric and magnetic fields normal to each other B1
- either* charged particle enters region normal to both fields
or correct B direction w.r.t. E for zero deflection B1
for no deflection, $v = E/B$ B1

(no credit if magnetic field region clearly not overlapping with electric field region)

(b) (i) $m = Bqr/v$ C1
 $= (640 \times 10^{-3} \times 1.6 \times 10^{-19} \times 6.2 \times 10^{-2}) / (9.6 \times 10^4)$ C1
 $= 6.61 \times 10^{-26} \text{ kg}$ C1
 $= (6.61 \times 10^{-26}) / (1.66 \times 10^{-27}) \text{ u}$
 $= 40 \text{ u}$ A1

(ii) $q/m \propto 1/r$ or m constant and $q \propto 1/r$ B1
 q/m for A is twice that for B B1
ions in path A have (same mass but) twice the charge (of ions in path B) B1

33.

(a) $F = BIL \sin \theta$ C1
 $= 2.6 \times 10^{-3} \times 5.4 \times 4.7 \times 10^{-2} \times \sin 34^\circ$
 $= 3.69 \times 10^{-4} \text{ N}$ A1
(allow 1 mark for use of $\cos 34^\circ$)

(b) peak current $= 1.7 \times \sqrt{2}$ C1
 $= 2.4 \text{ A}$

max. force $= 2.6 \times 10^{-3} \times 2.4 \times 4.7 \times 10^{-2} \times \sin 34^\circ$
 $= 1.64 \times 10^{-4} \text{ N}$ C1

variation $= 2 \times 1.64 \times 10^{-4}$
 $= 3.3 \times 10^{-4} \text{ N}$ A1

34.

(a) *either* charge exists in discrete and equal quantities B1
or multiples of elementary charge $/e/1.6 \times 10^{-19} \text{ C}$

(b) (i) force due to magnetic field must be upwards B1
B-field into the plane of the paper B1

(ii) sketch showing: deflection consistent with force in (b)(i) B1
reasonable curve B1

35.

- (a) (i) gravitational B1
- (ii) gravitational and electric B1
- (iii) magnetic and one other field given B1
magnetic, gravitational and electric B1
- (b) (i) out of (plane of) paper/page (*not* “upwards”) B1
- (ii) $B = mv / qr$ C1
 $= (3.32 \times 10^{-26} \times 7.6 \times 10^4) / (1.6 \times 10^{-19} \times 6.1 \times 10^{-2})$ C1
 $= 0.26 \text{ T}$ A1
- (c) sketch: semicircle with diameter $< 12.2 \text{ cm}$ B1

36.

- (a) (i) force = mg M1
along the direction of the field/of the motion A1
- (ii) no force B1
- (b) (i) force due to E -field downwards so force due to B -field upwards B1
into the plane of the paper B1
- (ii) force due to magnetic field = Bqv B1
force due to electric field = Eq B1
(*use of F_B and F_E not explained, allow 1/2*)
- forces are equal (and opposite) so $Bv = E$ or $Eq = Bqv$ so $E = Bv$ B1
- (c) sketch: smooth curved path M1
in ‘upward’ direction A1

37.

- (a) magnetic fields are equal in magnitude/strength/flux density M1
magnetic fields are opposite in direction M1
fields superpose/add/cancel to give zero/negligible resultant field A1
- (b) core causes increase in magnetic flux in the solenoid/induced poles in core B1
or field induced in core
changing flux threads/cuts the turns on the solenoid M1
(by Faraday’s law) an e.m.f. is induced in the solenoid A1
by Lenz’s law, this e.m.f. opposes the battery e.m.f. A1

38.

- (a) direction of force due to electric field opposite to force due to magnetic field
electric field is up the page B1
B1
- (b) force due to electric field = force due to magnetic field or $Eq = Bqv$ B1
 $E = Bv$ C1
 $= 9.7 \times 10^{-2} \times 1.6 \times 10^5$
 $= 1.6 (1.55) \times 10^4 \text{ V m}^{-1}$ A1
- (c) $q/m = v/Br$ C1
 $= 1.6 \times 10^5 / (9.7 \times 10^{-2} \times 4.0 \times 10^{-2})$ C1
 $= 4.1 (4.12) \times 10^7 \text{ C kg}^{-1}$ A1
- (d) (i) $m = (3 \times 1.60 \times 10^{-19}) / (4.12 \times 10^7)$ C1
 $m = 1.16 \times 10^{-26} / 1.66 \times 10^{-27}$
 $= 7(.0) \text{ u (allow 7.1 u)}$ A1
- (ii) 3 protons, 4 neutrons A1

39.

- (a) (i) change in flux linkage $= 40 \times (5.0 - 3.0) \times 10^{-6}$
 $= 8(.0) \times 10^{-5} \text{ Wb}$ A1
- (ii) time taken $= 8.0 \times 10^{-5} / 5.0 \times 10^{-4}$
 $= 0.16 \text{ (s)}$ C1
speed $= 3.0 \times 10^{-2} / 0.16$
 $= 0.19 (0.188) \text{ m s}^{-1}$ A1
- or
- $E = (\Delta\Phi / \Delta x) \times \text{speed}$
speed $= 5.0 \times 10^{-4} / (8.0 \times 10^{-5} / 3.0 \times 10^{-2})$ (C1)
 $= 0.19 (0.188) \text{ m s}^{-1}$ (A1)
- (b) a constant non-zero value of E from 0 to 3 cm and
a different constant non-zero value of E from 3 to 6 cm M1
 E from 3–6 cm has the opposite sign to and larger value than E from 0–3 cm A1

40.

(a) 0.10 mm B1

(b) $V_H = (0.13 \times 3.8) / (6.0 \times 10^{28} \times 0.10 \times 10^{-3} \times 1.60 \times 10^{-19})$ C1

$= 5.1 \times 10^{-7} \text{ V}$ A1

41.

(a) (non-uniform) magnetic flux in core is changing M1

induces (different) e.m.f. in (different parts of) the core A1

(eddy) currents form in the core M1

which give rise to heating A1

(b) as magnet falls, tube cuts magnetic flux M1

e.m.f./ (eddy) currents induced in metal/aluminium (tube) A1

(eddy) current heating of tube M1

with energy taken from falling magnet A1

or

(eddy) currents produce magnetic field (M1)

that opposes motion of magnet (A1)

so magnet B has acceleration $< g$

or

magnet B has smaller acceleration/reaches terminal speed A1

42.

(a) (i) number density of charge carriers/free electrons
or
number per unit volume of charge carriers/free electrons

B1

(ii) PX or QY or RZ B1

- (b) (i) V_H is inversely proportional to n B1
 for semiconductors, n is (much) smaller than for metals B1
 (ii) magnetic field would deflect holes and electrons in same direction B1
 (because) electrons are (–)ve, holes are (+)ve M1
 so V_H has opposite polarity/opposite sign A1

43.

- (a) iron rod changes flux (density)/field B1
 change of flux in coil Q causes induced e.m.f. B1
 (b) constant reading (either polarity) from time zero to near t_1 B1
 spike in one direction near t_1 clearly showing a larger voltage M1
 of opposite polarity A1
 zero reading from near t_1 to t_2 B1

44.

- (a) region (of space) where a force is experienced by a particle B1
 (b) (i) gravitational B1
 (ii) gravitational and electric B1
 (iii) gravitational, electric and magnetic B1
 (c) (i) force (always) normal to direction of motion M1
 (magnitude of) force constant
 or
 speed is constant/kinetic energy is constant M1
 magnetic force provides/is the centripetal force A1
 (ii) $mv^2/r = Bqv$ B1
 momentum or p or $mv = Bqr$ B1

45.

- (a) (induced) e.m.f. proportional to rate
of change of (magnetic) flux (linkage)

M1
A1

- (b) flux linkage = BAN

$$= \pi \times 10^{-3} \times 2.8 \times \pi \times (1.6 \times 10^{-2})^2 \times 85 = 6.0 \times 10^{-4} \text{ Wb}$$

B1

- (c) e.m.f. = $\Delta N\Phi / \Delta t$

$$= (6.0 \times 10^{-4} \times 2) / 0.30$$

C1

$$= 4.0 \text{ mV}$$

A1

- (d) sketch: $E = 0$ for $t = 0 \rightarrow 0.3 \text{ s}$, $0.6 \text{ s} \rightarrow 1.0 \text{ s}$, $1.6 \text{ s} \rightarrow 2.0 \text{ s}$

B1

$$E = 4 \text{ mV for } t = 0.3 \text{ s} \rightarrow 0.6 \text{ s (either polarity)}$$

B1

$$E = 2 \text{ mV for } t = 1.0 \text{ s} \rightarrow 1.6 \text{ s}$$

B1

with opposite polarity

B1

46.

- (a) (by Newton's third law) force on wire is up(wards)
by (Fleming's) left-hand rule/right-hand slap rule to give current
in direction left to right shown on diagram

M1
A1
A1

- (b) force $\propto$ current or $F = BIL$ or $B (= 0.080 / 6.0L) = 1 / 75L$

C1

$$\begin{aligned} \text{maximum current} &= 2.5 \times \sqrt{2} \\ &= 3.54 \text{ A} \end{aligned}$$

C1

$$\begin{aligned} \text{maximum force in one direction} &= (3.54 / 6.0) \times 0.080 \\ &= 0.047 \text{ N} \end{aligned}$$

C1

$$\text{difference } (= 2 \times 0.047) = 0.094 \text{ N}$$

or

force varies from 0.047 N upwards to 0.047 N downwards

A1

47.

8(a)	region (of space) where there is a force	M1
	produced by / on a magnet / magnetic pole / <u>moving</u> charge / current-carrying conductor	A1
8(b)(i)	out of (the plane of) the paper / page	B1
8(b)(ii)	the force on the particle is (always) perpendicular to the velocity / perpendicular to the direction of travel / towards the centre of path	B1
	no work is done by the force on the particle / there is no acceleration in the direction of the velocity / the acceleration is (always) perpendicular to the velocity	B1
8(b)(iii)	$F = Bqv$ or $F = mv^2 / r$	C1
	$mv^2 / (d/2) = Bqv$ so $d = 2mv / Bq$	A1
8(b)(iv)	time = distance / speed $T_{(F)} = \pi d / 2v$	C1
	$T_{(F)} = (\pi / 2v) \times (2mv / Bq)$ $T_{(F)} = \pi m / Bq$ and so $T_{(F)}$ independent of v	A1

48.

7(a)	(magnetic) force (always) normal to velocity / direction of motion	M1
	(magnitude of magnetic) force constant or speed is constant / kinetic energy is constant	M1
	so provides the centripetal force	A1
7(b)	increase in KE = loss in PE or $\frac{1}{2}mv^2 = qV$	M1
	$p = mv$ with algebra leading to $p = \sqrt{2mqV}$	A1
7(c)	$Bqv = mv^2 / r$ $mv = Bqr$ or $p = Bqr$	C1
	$(2 \times 9.11 \times 10^{-31} \times 1.60 \times 10^{-19} \times 120)^{1/2} = B \times 1.60 \times 10^{-19} \times 0.074$	C1
	$B = 5.0 \times 10^{-4} \text{ T}$	A1
7(d)	greater momentum	M1
	($p = Bqr$ and) so r increased	A1

49.

9(a)(i)	Hall voltage depends on thickness of slice	C1
	thinner slice, larger Hall voltage	A1
9(a)(ii)	Hall voltage depends on current in slice	B1
9(b)	sinusoidal wave, one cycle	B1
	at $\theta = 0$ and at $\theta = 360^\circ$, $V_H = V_{\text{MAX}}$	B1
	at $\theta = 180^\circ$, $V_H = -V_{\text{MAX}}$	B1

50.

8(a)	DERQ and CFSP	B1
8(b)(i)	force (on charge) due to magnetic field = force due to electric field or $Bqv = Eq$ or $v = E/B$	B1
	$E = V_H/d$	B1
	$V_H = Bvd$	B1
8(b)(ii)	use of $I = nAqv$ and $A = dt$	M1
	algebra clear leading to $V_H = BI/ntq$	A1
8(c)	(in metal,) n is very large	M1
	(therefore) V_H is small	A1

51.

8(a)(i)	DERQ and CFSP	B1
8(a)(ii)	charge carriers moving normal to (magnetic) field	B1
	<u>charge carriers</u> experience a <u>force</u> normal to I (and B)	B1
	charge build-up sets up electric field across the slice or build-up of charges results in a p.d. across the slice	B1
	charge stops building up/ V_H becomes constant when $F_B = F_E$	B1
8(b)	V_H inversely proportional to n /number density of charge carriers	B1
	number density of charge carriers (n) lower in semiconductors so V_H larger for semiconductor slice	B1
	or	
	V_H proportional to v /drift velocity	(B1)
	(for same current) drift velocity (v) higher in semiconductors so V_H larger for semiconductor slice	(B1)

52.

9(a)	region (of space)	B1
	where an object/particle experiences a force	B1
9(b)	electric and magnetic fields normal to each other	B1
	velocity of particle normal to both fields	B1
	forces (on particle) due to fields are in opposite directions	B1
	<u>forces are equal</u> for particles with a particular speed/for a selected speed/for speed given by $v = E(q)/B(q)$	B1
9(c)(i)	path labelled Q shown undeviated	B1
9(c)(ii)	reasonable curve in field and no 'kink' on entering, labelled V	B1
	deviated 'upwards'	B1

53.

9(a)(i)	PSYV <u>and</u> QRXW	B1
9(a)(ii)	electron moving in magnetic field deflected towards face QRXW	M1
	so face PSYV is more positive	A1
9(b)(i)	PV <i>or</i> SY <i>or</i> RX <i>or</i> QW	B1
9(b)(ii)	number of charge carriers per unit volume	B1
9(b)(iii)	negative and positive charge (carriers) would deflect in opposite directions	M1
	so no change in polarity	A1

54.

10(a)(i)	<i>either</i> product of flux density and area	M1
	direction of flux normal to area	A1
	<i>or</i> flux density $\times$ area $\times \sin \theta$	(M1)
	where θ is angle between direction of flux and area	(A1)
10(a)(ii)	(induced) e.m.f. proportional to rate	M1
	of change of (magnetic) flux linkage	A1
10(b)	e.m.f. $= \Delta(\phi N) / \Delta t$	C1
	$= (6.8 \times 10^{-6} \times 2 \times 3.5 \times 96) / (2.4 \times 10^{-3})$	
	$= 1.9 \text{ V}$	A1
10(c)	alternating	C1
	with same frequency as supply	A1

55.

9(a)	using Fleming's left-hand rule force on wire is upwards	B1
	by Newton's third law, force on magnet is downwards	B1
9(b)(i)	$F = BIL$	C1
	$= 3.7 \times 10^{-3} \times 5.1 \times 8.5 \times 10^{-2}$	A1
	$= 1.6 \times 10^{-3} \text{ N}$	
9(b)(ii)	$F = 1.6 \times 10^{-3} \text{ N}$	A1
9(c)	sketch: sinusoidal wave with two cycles	B1
	amplitude $2.3 \times 10^{-3} \text{ N}$	B1
	period 0.05 s	B1

56.

10(a)	induced <u>e.m.f.</u> proportional to rate	M1
	of <u>change</u> of (magnetic) <u>flux</u> (linkage)	A1
	<u>or</u> of <u>cutting</u> (magnetic) <u>flux</u>	
10(b)	current in coil produces flux	B1
	(by Faraday's law) changing flux induces e.m.f. in ring	B1
	current in ring causes field (around ring)	B1
	(by Lenz's law) field around ring opposes field around coil	B1

57.

8(a)	electric and magnetic fields at right-angles to one another (<i>may be shown on a clearly labelled diagram</i>)	B1
	particle enters fields (with velocity) normal to the (two) fields (<i>may be shown on a clearly labelled diagram</i>)	B1
	no deviation for particles with selected velocity	B1
8(b)	magnetic force equals/is the centripetal force	C1
	$Bqv = mv^2/r$	C1
	$M = Bqr/v$	M1
	$= (94 \times 10^{-3} \times 1.6 \times 10^{-19} \times 0.075) / (3.4 \times 10^4)$	
	<u>division</u> by 1.66×10^{-27} shown, to give $m = 20 \text{ u}$	A1
8(c)	sketch: semicircle clear (in same direction)	B1
	with larger radius	B1

58.

9(a)	(magnetic) flux density $\times$ area	B1
	magnetic flux density normal to area or reference to cross-sectional area or $\times \sin$ (angle between B and A)	B1
	$\times$ number of turns on coil	B1
9(b)	e.m.f. = BAN/t or e.m.f. = rate of change of flux <u>linkage</u>	C1
	$= (7.5 \times 10^{-3} \times \pi \times \{1.2 \times 10^{-2}\}^2 \times 160) / 0.15$	A1
	$= 3.6 \times 10^{-3} \text{ V}$	
9(c)	sketch: zero for 0–0.10 s, 0.25–0.35 s, and 0.425–0.55 s, and non-zero outside these ranges	B1
	two horizontal steps, with zero voltage either side	B1
	with same polarity	B1
	correct values (1st step 3.6 mV and 2nd step 7.2 mV)	B1

59.

8(a)	region where there is a force	M1
	experienced by a current-carrying conductor/moving charge/(permanent) magnet	A1
8(b)(i)	single path, deflection in 'upward' direction	B1
	acceptable circular arc in whole field	B1
	no 'kinks' at start or end of curvature, and straight outside region of field	B1
8(b)(ii)	force (on particle) is normal to velocity/direction of motion/direction of speed	B1
8(c)	magnetic force provides/is the centripetal force	B1
	$Bqv = mv^2/r$ or $r = mv/Bq$	C1
	(if q is doubled), new speed = $2v$	A1

60.

9(a)	(induced) e.m.f. proportional/equal to <u>rate</u>	M1
	of change of (magnetic) flux (linkage)	A1
9(b)(i)	induced e.m.f. = $(\Delta B)AN/\Delta t$	C1
	= $(2 \times 0.19 \times 1.5 \times 10^{-4} \times 120)/0.13$	
	= 0.053 V	A1
9(b)(ii)	reading on voltmeter connected to coil C/V: 0 0.053 0 (all three values required)	A1
	reading on voltmeter connected to Hall probe/V: zero in middle column	B1
	final column correct sign (negative)	B1
	final column correct magnitude (0.20)	B1

61.

8(a)	force per unit current	B1
	force per unit length (of wire)	B1
	current normal to (magnetic) field	B1
8(b)(i)	forces (on PQ and RS) are horizontal	B1
	(hence they create) no moment about the pivot	B1
	or	
	forces (on PQ and RS) are equal and opposite	(B1)
	(hence there is) no <u>net</u> force (on the two sections)	(B1)
8(b)(ii)	realisation of the need to apply moments	C1
	$BILx = mgy$	C1
	$B \times 2.7 \times 1.2 \times 10^{-2} \times 7.5 = 45 \times 10^{-6} \times 9.81 \times 8.8$	
	$B = 1.6 \times 10^{-2} \text{ T}$	A1

62.

9(a)	$0 \rightarrow t_1$ horizontal straight line at non-zero value of V_H and $t_3 \rightarrow t_4$ horizontal straight line at different non-zero V_H	B1
	$t_1 \rightarrow t_3$ straight diagonal line with negative gradient and graph line starts at $(0, V_0)$ and ends at $(t_4, -2V_0)$	B1
9(b)	$E = 0$ for $0 \rightarrow t_1$ and $t_3 \rightarrow t_4$	B1
	E is non-zero at all points between $t_1 \rightarrow t_3$	M1
	E has constant magnitude between $t_1 \rightarrow t_3$	A1

63.

8(a)(i)	Either Newton's third law or equal and opposite forces	B1
	force on magnet is upwards	B1
	so force on wire downwards	B1
8(a)(ii)	using (Fleming's) left-hand rule	M1
	current from B to A	A1
8(b)	sinusoidal wave with at least 1 cycle	B1
	peaks at +6.4 mN and -6.4 mN	B1
	time period 25 ms	B1

64.

10(a)	single straight line along full length of solenoid	B1
	at least two more parallel lines along full length of solenoid	B1
	correct direction – right to left	B1
10(b)	(induced) e.m.f. proportional / equal to <u>rate</u>	M1
	of change of (magnetic) flux (linkage)	A1
10(c)	increasing current causes increasing flux	B1
	increasing flux induces e.m.f. in coil	B1
	(induced) e.m.f. opposes growth of current	B1

65.

8(a)(i)	region where a force is exerted on: a magnetic pole or a moving charge or a current-carrying wire	B1
8(a)(ii)	arrow on axis of solenoid pointing downwards labelled P	B1
8(b)(i)	<u>direction</u> of induced e.m.f./current	M1
	(tends to) oppose the change causing it	A1
8(b)(ii)	magnetic field in solenoid is increasing	B1
	field in coil in opposite direction to oppose increase	B1
	arrow inside or just above small coil pointing in opposite direction to P	B1
8(c)	e.m.f. = $N\Delta\phi / \Delta t$	C1
	= $(75 \times 1.4 \times 10^{-3} \times 2 \times 7.0 \times 10^{-4}) / 0.12$	C1
	= $1.2 \times 10^{-3} \text{ V}$	A1

66.

8(a)(i)	$v_N = 3.4 \times 10^7 \times \sin 30^\circ$ $= 1.7 \times 10^7 \text{ m s}^{-1}$	A1
8(a)(ii)	$mv^2 / r = Bqv$ or $r = mv / Bq$	C1
	$r = (9.11 \times 10^{-31} \times 1.7 \times 10^7) / (3.2 \times 10^{-3} \times 1.60 \times 10^{-19})$	C1
	= 0.030 m	A1
8(b)	zero	B1
8(c)	helix/coil	B1

67.

8(a)	concentric circles (around the wire)	M1
	at least 3 circles shown, all with increasing separation	A1
	direction anticlockwise	B1
8(b)(i)	$B = (4\pi \times 10^{-7} \times 6.2) / (2\pi \times 3.1 \times 10^{-2})$	C1
	$= 4.0 \times 10^{-5} \text{ T}$	A1
8(b)(ii)	$F = BIL$ or $F/L = BI$	C1
	$F/L = 4.0 \times 10^{-5} \times 8.5$	A1
	$= 3.4 \times 10^{-4} \text{ N m}^{-1}$	
8(c)	correct application of Newton's 3rd law to the forces or F/L is proportional to the product of the two currents	M1
	so same magnitude	A1

68.

8(a)	magnitude: (force =) Bqv	B1
	direction: $P \rightarrow Q$ or $E \rightarrow F$ or $S \rightarrow R$ or $H \rightarrow G$	B1
8(b)(i)	EHSP and FGRQ	B1
8(b)(ii)	PE or QF or RG or SH	B1
8(c)(i)	any one correct starting point from: (mass of 1 atom =) $27 \times 1.66 \times 10^{-27}$ (amount of substance per unit volume =) $2.7 / 27$ 27 g (of substance) contains 6.02×10^{23} atoms (2.7 g mass contains) 0.1 mol (1 cm^3 volume contains) 0.1 mol (1 m^3 volume contains) 10^5 mol	C1
	$n = (2.7 \times 10^3) / (27 \times 1.66 \times 10^{-27}) = 6.0 \times 10^{28}$ or $n = (2.7 / 27) \times 10^6 \times 6.02 \times 10^{23} = 6.0 \times 10^{28}$	A1
8(c)(ii)	$V_H = (0.15 \times 4.6) / (6.0 \times 10^{28} \times 0.090 \times 10^{-3} \times 1.60 \times 10^{-19})$	C1
	$= 8.0 \times 10^{-7} \text{ V}$	A1

69.

8(a)	a region where a magnet / magnetic material / moving charge / current carrying conductor experiences a force	B1
8(b)	$B = F / Il$ e.g. $= 9 \times 10^{-3} / (5.0 \times 0.045)$	C1
	$= 0.040 \text{ T}$	A1
8(c)(i)	force is (always) perpendicular to the velocity / direction of motion	B1
	magnetic force provides the centripetal force or force perpendicular to motion causes circular motion	B1
	magnitude of force (due to the magnetic field) is constant or no work done by force or the force does not change the speed	B1
8(c)(ii)	Applying the list rule, any 2 from: accelerating p.d. radius of path / radius of semicircle magnetic flux density	B2

70.

8(a)	magnetic field normal to current	B1
	newton per ampere	B1
	newton per metre	B1
8(b)(i)	current in wire QL gives rise to a force or wire QL is perpendicular to the magnetic field	B1
	force on wire QL is vertical	B1
	force does not act through the pivot	B1
8(b)(ii)	forces act through the same line or forces are horizontal	B1
	forces are equal (in magnitude) and opposite (in direction)	B1
8(c)(i)	change = $mg \times (\Delta)L$	C1
	$= 1.3 \times 10^{-4} \times 9.81 \times 2.6 \times 10^{-2} = 3.3 \times 10^{-5} \text{ N m}^{-1}$	A1
8(c)(ii)	change = $B \times (\Delta)I \times L \times x$	C1
	$3.3 \times 10^{-5} = B \times 1.2 \times 0.85 \times 10^{-2} \times 5.6 \times 10^{-2}$	C1
	$B = 0.058 \text{ T}$	A1

71.

9(a)(i)	e.m.f. = $(\Delta)B \times AN / t$	C1
	$= 45 \times 10^{-3} \times \pi \times (1.8 \times 10^{-2})^2 \times 350 / 0.20 = 0.080 \text{ V}$	A1
9(a)(ii)	0 to 0.2 s: straight horizontal line at 0.080 V or -0.080 V	B1
	0.2 s to 0.4 s: zero	B1
	0.4 s to 0.8 s: straight horizontal line at 0.040 V or -0.040 V	B1
	opposite polarity to 0 to 0.2 s line	B1
9(b)	either disc cuts flux lines (of the magnet) or there is a changing flux in the disc	B1
	(by Faraday's law) e.m.f. is induced in the disc	B1
	e.m.f. causes (eddy) currents in the disc	B1
	current in the magnetic field (of the magnet) causes force on disc	B1

72.

9(a)(i)	force is downwards/down the page or current is (right) to left	B1
	by left-hand rule, field is into plane of paper	B1
9(a)(ii)	magnetic force provides the centripetal force	C1
	$Bqv = mv^2 / r$	C1
	$v = Bqr / m$	A1
	$= (8.0 \times 10^{-4} \times 1.60 \times 10^{-19} \times 6.4 \times 10^{-2}) / (9.11 \times 10^{-31})$ $= 9.0 \times 10^6 \text{ m s}^{-1}$	

9(b)(i)	arrow showing field direction down the page	B1
9(b)(ii)	$Bqv = Eq$ or $v = E / B$	C1
	$E = 9.0 \times 10^6 \times 8.0 \times 10^{-4}$ $= 7.2 \times 10^3 \text{ N C}^{-1}$	A1
9(c)	straight line/undeviated	B1
	condition for no deflection depends only on v or condition for no deflection does not depend on m or q	B1

73.

8(a)(i)	downwards	B1
8(a)(ii)	PQRS and JKLM	B1
8(b)	(as charge separates) an electric field is created (between opposite faces)	B1
	(maximum value is reached when) electric force (on electron) is equal and opposite to magnetic force (on electron)	B1
8(c)	$V_H = BI / ntq$ $= (4.6 \times 10^{-3} \times 6.3 \times 10^{-4}) / (1.3 \times 10^{29} \times 0.10 \times 10^{-3} \times 1.60 \times 10^{-19})$	C1
	$= 1.4 \times 10^{-12} \text{ V}$	A1
8(d)	semiconductors have a (much) smaller value for n	B1
	V_H for semiconductors is (much) larger so more easily measured	B1

74.

9(a)(i)	(induced) e.m.f. (directly) proportional to rate	M1
	of change of magnetic flux (linkage)	A1
9(a)(ii)	e.m.f. = 0 apart from thin pulses at t_1 and t_2	B1
	rectangular pulses centred on t_1 and t_2 , of widths 2 small squares and 1 small square respectively	B1
	e.m.fs. at t_1 and t_2 have opposite polarities	B1
	magnitude of e.m.f. at t_2 double the magnitude of e.m.f. at t_1	B1
9(b)	V_H shown as zero before ($t_1 - 2$ squares) and after ($t_2 + 2$ squares) and rises to a constant non-zero value between t_1 and t_2	M1
	change at t_1 shown as 2 small squares wide and change at t_2 shown as 1 small square wide	A1

75.

10(a)	concentric circles centred on the wire	B1
	separation of lines increasing with distance from wire	B1
	arrows show anti-clockwise direction	B1
10(b)(i)	current in (each) wire creates a magnetic field (at the other wire)	B1
	current (in wire) at 90° to field causes force	B1
10(b)(ii)	force on each wire towards other wire/attractive	B1
10(c)	Newton's third law pair of forces so yes (forces are equal) or force proportional to product of both currents so yes (forces are equal)	B1

76.

8(a)(i)	at least one anticlockwise arrow and no clockwise arrows	B1
8(a)(ii)	(force is to the) left	B1
8(a)(iii)	force is the same	B1
	Newton's third law (of motion) or force depends on the product of the two currents	B1

77.

9(a)	(magnetic) flux density $\times$ area $\times$ number of turns	M1
	area is perpendicular to (magnetic) field	A1
9(b)	use of $t = 1.2$ s	C1
	$\varepsilon = \frac{\Delta BAN}{\Delta t}$ $= \frac{0.250 \times \pi \times 0.030^2 \times 540}{1.2}$	C1
	$= 0.32$ V	A1
9(c)(i)	light damping	B1
9(c)(ii)	sheet cuts (magnetic) flux and causes induced emf	B1
	(induced) emf causes (eddy) currents (in sheet)	B1
	either currents (in sheet) cause resistive force or currents (in sheet) dissipate energy	B1
	smaller currents in Y or larger currents in X, so dashed line is X	B1

78.

9(a)	region where there is a force exerted on	M1
	a current-carrying conductor or a moving charge or a magnetic material/magnetic pole	A1
9(b)(i)	face PSWV shaded	B1
9(b)(ii)	accumulating electrons cause an electric field (between the faces)	B1
	force due to electric field opposes force due to magnetic field	B1
	accumulation stops when magnetic force equals electric force	B1
9(c)(i)	number density of charge carriers	B1
9(c)(ii)	PV or QT or SW	B1
9(d)	(for semiconductor,) n is (much) smaller so V_H (much) larger	B1

79.

10(a)	direction of (induced) e.m.f.	M1
	is such as to oppose the <u>change</u> causing it	A1
10(b)	ring cuts (magnetic) flux and causes induced e.m.f. in ring	B1
	(induced) e.m.f. causes (eddy/induced) currents (in ring)	B1
	currents (in ring) cause magnetic field (around ring)	M1
	two fields interact to cause resistive/opposing force	A1
	or	
	current (in ring) is in a magnetic field	(M1)
	which causes resistive force	(A1)
	or	
	currents (in ring) dissipate thermal energy	(M1)
	(thermal) energy comes from energy of oscillations	(A1)
10(c)	current cannot pass all the way around the ring	B1
	(induced) currents smaller	B1
	smaller resistive force (so more oscillations)	B1
	or smaller <u>rate</u> of dissipation of energy (so more oscillations)	

80.

8(a)	- force per unit length - force per unit current - length/current perpendicular to field <i>1 mark for any two points, 2 marks for all three points</i>	B2
8(b)	change in potential energy = change in kinetic energy or $qV = \frac{1}{2}mv^2$	B1
	$v = \sqrt{(2qV/m)}$	A1
8(c)(i)	magnetic force = centripetal force or $Bqv = mv^2/r$	M1
	clear substitution of expression for v and correct algebra leading to $q/m = 2V/B^2r^2$	A1
8(c)(ii)	$q/m = (2 \times 230) / [(0.38 \times 10^{-3})^2 \times 0.14^2]$	C1
	$= 1.6 \times 10^{11} \text{ C kg}^{-1}$	A1
8(c)(iii)	(for α -particle,) q/m is (much) smaller	B1
	r would be <u>much</u> larger	B1

81.

9(a)	(particle is) stationary/not moving	B1
	(particle is) moving parallel to the (magnetic) field	B1
9(b)	magnetic field around each coil is circular or each coil is normal to magnetic field due to adjacent coils	B1
	current in coil interacts with (magnetic) field to exert force (on coil)	B1
	force is normal to both coil and magnetic field or force parallel to axis (of coil)	B1
	forces between coils are attractive so spring contracts	B1
9(c)	(oscillating) coils cut magnetic flux or as separation of coils changes, magnetic flux changes	B1
	cutting flux causes induced e.m.f. in coils	B1
	<u>changing</u> (induced) e.m.f. causes changing current (in coil)	B1

82.

8(a)	newton per ampere per metre	M1
	where current/wire is perpendicular to magnetic field	A1
8(b)(i)	$F = BIL \sin \theta$	C1
	$B = 1.0 / (5.0 \times 0.060 \times \sin 50^\circ)$ $= 4.4 \text{ mT}$	A1
8(b)(ii)	(from Fleming's left-hand rule) force on wire is upwards, so reading decreases	B1
8(b)(iii)	frame will rotate (so that PQ becomes perpendicular to the field)	B1

83.

8(a)(i)	arrow from Q pointing downwards, labelled B	B1
8(a)(ii)	arrow from Q pointing towards P, labelled F	B1
8(b)(i)	force is proportional to product of both currents (I and $2I$) or Newton's third law	B1
	forces are equal	B1
8(b)(ii)	opposite	B1

84.

6(a)	less in smaller solenoid	B1
6(b)	greater in smaller solenoid	B1
6(c)(i)	<u>direction</u> of (induced) e.m.f.	M1
	such as to (produce effects that) oppose the <u>change</u> that caused it	A1
6(c)(ii)	change of flux (linkage) in smaller solenoid induces e.m.f. in smaller solenoid	B1
	(induced) current in smaller solenoid causes field around it	B1
	the two fields (interact to) create an attractive force	B1

85.

6(a)	product of (magnetic) flux density and area	M1
	where area is perpendicular to the (magnetic) field	A1
6(b)(i)	$N\Phi = BAN$	C1
	$= 400 \times 10^{-3} \times 0.12^2 \times 8$	C1
	$= 0.046 \text{ Wb}$	A1
6(b)(ii)	(line is a) straight line	B1
6(b)(iii)	(induced) e.m.f. = rate of change of flux linkage	C1
	e.m.f. $= N\Phi / t$	A1
	$= 0.046 / 0.60$	
	$= 0.077 \text{ V}$	
6(c)	(induced e.m.f. causes) current flow (in the coil)	B1
	either	
	current (in magnetic field) causes forces to act on the coil	B1
	(opposite sides of) coil forced inwards	B1
	or	
	current causes dissipation of energy in the resistance of the coil	(B1)
	temperature of the coil rises	(B1)

86.

6(a)	there must be a current (in the wire)	B1
	(wire) must be at a non-zero angle to the magnetic field	B1
6(b)(i)	arrow from X pointing horizontally to the left	B1
	arrow from Y pointing diagonally upwards and to the left at about 45°	B1
	arrow from Z pointing horizontally to the right	B1
6(b)(ii)	(flux densities at W and X are approximately) equal	B1
	(flux density at) Y greater than (flux density at) Z	B1
6(c)	current in wire creates magnetic field around wire	B1
	(each) wire sits in the magnetic field created by the other	B1
	(for each wire,) current / wire is perpendicular to magnetic field (due to other wire), (so) experiences a (magnetic) force	B1

87.

7(a)	induced e.m.f. is (directly) proportional to rate	M1
	of change of (magnetic) flux (linkage)	A1
7(b)	V_2 stepped, all at non-zero values, between $t = 0$ and $t = 0.40 \text{ s}$	B1
	V_2 shown with same non-zero magnitude up to $t = 0.15 \text{ s}$ and after $t = 0.25 \text{ s}$ but with a different magnitude between these times	B1
	V_2 shown with a magnitude between $t = 0.15 \text{ s}$ and $t = 0.25 \text{ s}$ that is three times the magnitude before $t = 0.15 \text{ s}$ and after $t = 0.25 \text{ s}$	B1
	V_2 shown with same sign up to $t = 0.15 \text{ s}$ and after $t = 0.25 \text{ s}$, and opposite sign in between	B1

7(c)(i)	changing current in coil causes changing (magnetic) field or changing (magnetic) flux causes induced e.m.f. in ring	B1
	induced e.m.f. in ring causes current in ring	B1
	(magnetic) field due to (induced) current in ring interacts with (coil's) field to cause upwards force (on ring) or (induced) current in ring perpendicular to (coil's magnetic) field causes upwards force (on ring)	B1
7(c)(ii)	both magnetic fields reverse direction so ring still jumps up or current (in ring) and (coil's) field both reverse so ring still jumps up	B1

88.

6(a)(i)	PQRS and WXYZ	B1
6(a)(ii)	force on charge carriers is perpendicular to both (magnetic) field and current	B1
	as charge carriers are deflected to one side, an electric field is set up	B1
	(steady V_H when) electric and magnetic forces on charge carriers are equal (and opposite)	B1
6(b)(i)	n : number density of charge carriers	B1
	t : distance PW (or SZ or QX or RY)	B1
	q : charge on each charge carrier	B1
6(b)(ii)	V_H inversely proportional to t	B1
	(so t needs to be small for) V_H to be large enough to measure	B1

89.

7(a)	force per unit current	M1
	force per unit length	M1
	current / wire is perpendicular to (magnetic) field (lines)	A1
7(b)(i)	current (in coil) is perpendicular to magnetic field (so force on wire)	B1
	force (on wire) is perpendicular to current and field (so is vertical) or current and field are both horizontal (so force is vertical)	B1
7(b)(ii)	$NBIL = mg$	C1
	$B = (2.16 \times 10^{-3} \times 9.81) / (40 \times 3.94 \times 0.0300)$	C1
	$= 4.48 \times 10^{-3} \text{ T}$	A1
7(b)(iii)	(magnetic) forces (on balance and newton meter) are (equal and) opposite	B1
	reading $= 0.563 - (2.16 \times 10^{-3} \times 9.81)$ $= 0.542 \text{ N}$	A1

90.

8(a)	direction of induced e.m.f.	M1
	such as to (produce effects that) oppose the change that caused it	A1
8(b)(i)	$X = 0.85 \text{ A}$	A1
	$Y = 2\pi / 0.040$	C1
	$= 160 \text{ rad s}^{-1}$	A1
8(b)(ii)	two cycles of a sinusoidal curve with a period of 0.040 s	B1
	correct phase (i.e. V_2 max / min at $t = 0, 0.02, 0.04, 0.06$ and 0.08 s , and V_2 zero at $t = 0.01, 0.03, 0.05, 0.07 \text{ s}$)	B1
	maximum / minimum V_2 shown (consistently) at $\pm 6.5 \text{ V}$	B1
8(b)(iii)	(magnitude of) V_2 is proportional to rate of change of (magnetic) flux	B1
	V_2 is proportional to <u>gradient</u> of I_1-t curve V_2 has maximum magnitude when I_1-t curve is steepest V_2 is zero when I_1-t curve is horizontal / a maximum or minimum V_2 changes sign when sign of gradient of I_1-t curve changes Any two points, 1 mark each	B2

91.

6(a)	it is zero when (plane of) probe is parallel to the (magnetic) field (lines)	B1
	it is maximum when (plane of) probe is perpendicular to (magnetic) field (lines)	B1
6(b)(i)	number density of charge carriers	B1
6(b)(ii)	smaller value of n so greater Hall voltage / V_H	B1
6(c)	(36 mV corresponds to) 48 mT	C1
	use of 1.4 s or (8.6 – 7.2) s	C1
	$E = \Delta BAN / \Delta t$	C1
	$= \frac{48 \times 10^{-3} \times 0.018^2 \times \pi \times 780}{1.4}$ $= 0.027 \text{ V}$	A1

92.

6(a)	a region where a force acts on	M1
	a current-carrying conductor	A1
	or a <u>moving</u> charge	
	or a magnetic material / magnetic pole	
6(b)	concentric circles around the wire	B1
	spacing between circles increases with distance from wire	B1
	arrows showing direction of field is clockwise	B1
6(c)(i)	$F = BIL$	C1
	force per unit length $= BI$	A1
	$= 2.6 \times 10^{-3} \times 5.0$	
	$= 0.013 \text{ N m}^{-1}$	
6(c)(ii)	to the right	B1
6(c)(iii)	force (per unit length) has the same magnitude due to Newton's 3rd law	B1
	$0.013 = 1.5 \times 10^{-3} \times I$ current = 8.7 A	A1

93.

6(a)(i)	product of (magnetic) flux density and area	M1
	area perpendicular to the (magnetic) field	A1
6(a)(ii)	flux = $B \times \pi r^2$	C1
	= $0.17 \times \pi \times 0.36^2$	
	= $6.9 \times 10^{-2} \text{ Wb}$	A1
6(b)	time for one revolution = $1 / 25 \text{ s}$	C1
	e.m.f. = rate of cutting flux or $\Delta \Phi / \Delta t$	C1
	= 0.069×25	A1
	= 1.7 V	
6(c)	current (in disc) is perpendicular to magnetic field or current causes force to act on disc	B1
	force opposes rotation of disc	B1
	left-hand rule indicates current is from rim to axle	B1

94.

6(a)	- force per unit length - force per unit current - length / current perpendicular to field <i>1 mark for any two points, 2 marks for all three points</i>	B2
6(b)(i)	into the page	B1
6(b)(ii)	$F = Bqv$	C1
	= $4.8 \times 10^{-3} \times 1.6 \times 10^{-19} \times 1.7 \times 10^7 = 1.3 \times 10^{-14} \text{ N}$	A1
6(b)(iii)	arrow at point X pointing down the page	B1
6(b)(iv)	$F = mv^2 / r$	C1
	$1.3 \times 10^{-14} = (9.11 \times 10^{-31}) \times (1.7 \times 10^7)^2 / r$	C1
	($r = 0.020 \text{ m}$)	A1
	$d = 2r$ $d = 0.040 \text{ m}$	
6(c)	path shows upwards deflection such that the curvature is always anticlockwise within the field	B1
	circular path with larger radius	B1
	line enters field at X and leaves field at distance $2d$ vertically from X	B1

95.

7(a)(i)	$V_H = BI / ntq$ $= (4.0 \times 10^{-6} \times 5.4) / (1.5 \times 10^{16} \times 1.8 \times 10^{-3} \times 1.60 \times 10^{-19}) = 5.0 \text{ V}$	A1
7(a)(ii)	sketch: straight diagonal line from (0, 0) to $t = 0.020 \text{ s}$ and straight diagonal line between two non-zero V_H values of same sign from $t = 0.040$ to 0.050 s	B1
	horizontal straight line at $V_H = 5.0 \text{ V}$ from $t = 0.020$ to 0.040 s	B1
	horizontal straight line at $V_H = 2.5 \text{ V}$ from $t = 0.050$ to 0.080 s	B1
7(b)(i)	e.m.f. = rate of change of (magnetic) flux (linkage)	C1
	$E = NA \Delta B / \Delta t$ or $E = NA \times \text{gradient (at } t = 0.010 \text{ s)}$	C1
	$E = 3000 \times 3.4 \times 10^{-4} \times (4.0 \times 10^{-6}) / (0.020) = 2.0 \times 10^{-4} \text{ V}$	A1
7(b)(ii)	sketch: line showing non-zero E from $t = 0$ to $t = 0.020 \text{ s}$ and from $t = 0.040 \text{ s}$ to $t = 0.050 \text{ s}$, and $E = 0$ at all other times	B1
	'top hats' showing constant non-zero E from $t = 0$ to $t = 0.020 \text{ s}$ and from $t = 0.040 \text{ s}$ to $t = 0.050 \text{ s}$	B1
	magnitude of E shown as $2.0 \times 10^{-4} \text{ V}$ in both non-zero sections	B1
	sign of E in the $t = 0$ to $t = 0.020 \text{ s}$ region opposite to the sign of E in the $t = 0.040 \text{ s}$ to $t = 0.050 \text{ s}$ region	B1